

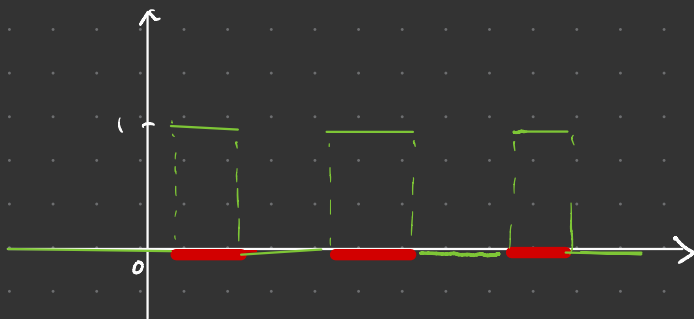

$E\{X \cdot E(Y|Z) | Z\}$ "Take Conditional Exp. given Z of the random variable $X \cdot E(Y|Z)$ "
 $= E(Y|Z) E[X|Z]$
 Conditional on Z , $E(Y|Z)$ can be taken to be constant

$$E[Y | X=x] = \int_{-\infty}^{\infty} y \cdot f(y|x) dy, \quad f(y|x) = \frac{f(x,y)}{f(x)}$$

A is a set of real numbers.

$$I_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

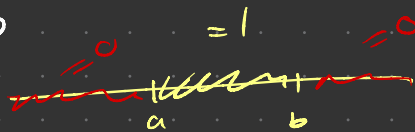
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of real numbers



with indicator fns. we can describe the support of a distribution.

Example: Uniform distribution over $[a, b]$ $a < b$

↳ Pdf: $\frac{1}{b-a} \cdot 1_{[a,b]}(x)$



$$\int_{-\infty}^{\infty} \frac{1}{b-a} 1_{[a,b]}(x) dx = \int_a^b \frac{1}{b-a} dx + \int_{-\infty}^a 0 \cdot \frac{1}{b-a} dx + \int_b^{\infty} 0 \cdot \frac{1}{b-a} dx$$

$$E^*(Y|X) = \alpha + \beta X$$

$\downarrow$ $\downarrow \rightarrow$

$$\beta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

$$\alpha = E(Y) - \beta E(X)$$

$\downarrow$ with data
 you estimate them
 $(\hat{\alpha}, \hat{\beta})$